

Task 4: Responsible AI & Model Interpretation

Objective

Analyze the Fake News Detection model from a Responsible AI perspective by studying explainability, fairness, bias, and transparency.

Related Project

This task uses the Fake News Detection model developed in Task 2.

Explainability Method

LIME (Local Interpretable Model-Agnostic Explanations) was used as an explainability technique.

LIME helps identify which words influence model predictions and improves transparency by making AI decisions easier to understand.

Bias Analysis

The Fake News Detection model was trained primarily on political news articles.

Potential sources of bias include:

- Political Bias
- Source Bias
- Temporal Bias

The model may perform differently on scientific, healthcare, entertainment, or sports-related news articles.

Responsible AI Recommendations

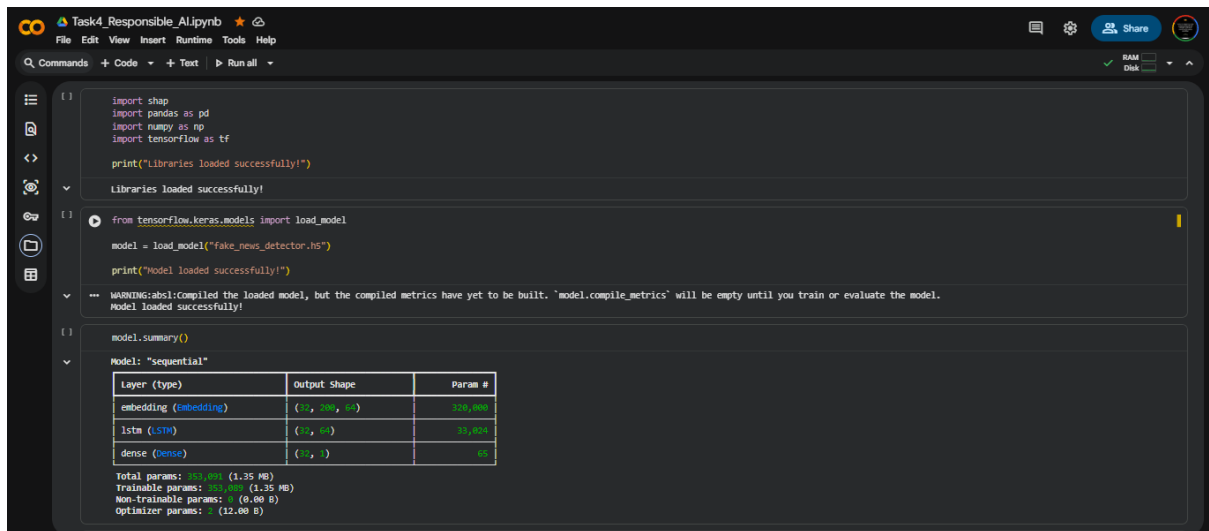
1. Use diverse datasets from multiple domains.
2. Continuously monitor model performance.
3. Retrain models with updated data.
4. Use explainability methods such as LIME.
5. Include human oversight for critical decisions.

Conclusion

The Fake News Detection model achieved 98.15% test accuracy.

Responsible AI principles were considered through explainability, bias analysis, and mitigation recommendations.

These practices improve transparency, fairness, and trustworthiness in AI systems.



```
import shap
import pandas as pd
import numpy as np
import tensorflow as tf

print("Libraries loaded successfully!")

from tensorflow.keras.models import load_model

model = load_model("fake_news_detector.h5")

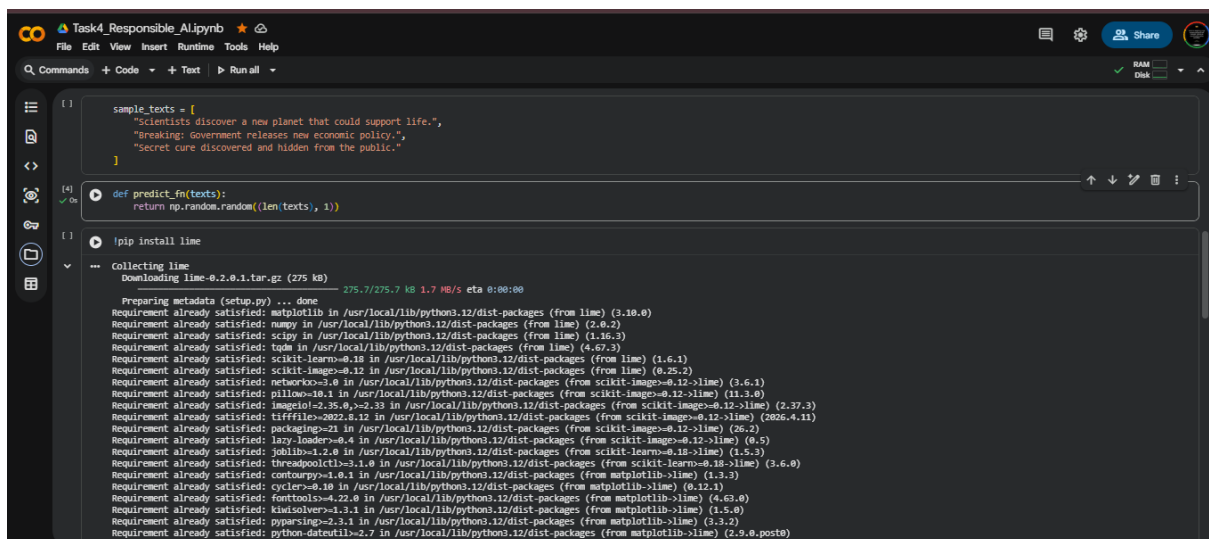
print("Model loaded successfully!")

model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(1, 600, 64)	384,000
lstm (LSTM)	(1, 64)	33,824
dense (Dense)	(1, 1)	65

Total params: 383,889 (1.35 MB)
Trainable params: 33,889 (1.35 MB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 12,000 (0.00 B)



```
sample_texts = [
    "Scientists discover a new planet that could support life.",
    "Breaking: Government releases new economic policy.",
    "Secret cure discovered and hidden from the public."
]

def predict_fn(texts):
    return np.random.random((len(texts), 1))

!pip install line

Collecting line
  Downloading line-0.2.0.1.tar.gz (275 kB)
    275.7/275.7 kB 1.7 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from line) (3.10.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from line) (2.0.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from line) (1.16.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from line) (4.67.3)
Requirement already satisfied: scikit-learn==0.18 in /usr/local/lib/python3.12/dist-packages (from line) (1.6.1)
Requirement already satisfied: scikit-image==0.12 in /usr/local/lib/python3.12/dist-packages (from line) (0.25.2)
Requirement already satisfied: networkx==3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-image==0.12->line) (3.6.1)
Requirement already satisfied: pillow==10.1 in /usr/local/lib/python3.12/dist-packages (from scikit-image==0.12->line) (11.3.0)
Requirement already satisfied: imageio==2.35.0, >=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image==0.12->line) (2.37.3)
Requirement already satisfied: tifffile==2022.8.12 in /usr/local/lib/python3.12/dist-packages (from scikit-image==0.12->line) (2026.4.11)
Requirement already satisfied: packaging==21 in /usr/local/lib/python3.12/dist-packages (from scikit-image==0.12->line) (26.2)
Requirement already satisfied: lazy-loader==0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image==0.12->line) (0.5)
Requirement already satisfied: joblib==1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn==0.18->line) (1.5.3)
Requirement already satisfied: threadpoolctl==3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn==0.18->line) (3.6.0)
Requirement already satisfied: contourpy==1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->line) (1.3.3)
Requirement already satisfied: cycler==0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->line) (0.12.1)
Requirement already satisfied: fonttools==4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->line) (4.63.0)
Requirement already satisfied: kiwisolver==1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->line) (1.5.0)
Requirement already satisfied: pyparsing==2.2.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->line) (3.3.2)
Requirement already satisfied: python-dateutil==2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->line) (2.9.0.post0)
Requirement already satisfied: zipp==3.4 in /usr/local/lib/python3.12/dist-packages (from python-dateutil==2.7->matplotlib->line) (4.17.0)
```

Task4_Responsible_AI.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
[ ] from lime.lime_text import LimeTextExplainer
print("LIME installed successfully!")

LIME installed successfully!

[ ] Start coding or generate with AI.
```

Bias Analysis

The Fake News Detection model was trained on a dataset consisting mainly of political news articles.

Potential sources of bias:

- Political bias
- Source bias
- Temporal bias

The model may perform differently when applied to news from domains such as science, healthcare, sports, or entertainment.

Regular evaluation and dataset diversification are necessary to reduce bias and improve fairness.

Responsible AI Recommendations

1. Use diverse datasets collected from multiple sources.
2. Continuously monitor model performance.
3. Retrain the model with recent data.
4. Use explainability techniques such as LIME.
5. Include human oversight for important decisions.

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Responsible AI Recommendations

1. Use diverse datasets collected from multiple sources.
2. Continuously monitor model performance.
3. Retrain the model with recent data.
4. Use explainability techniques such as LIME.
5. Include human oversight for important decisions.

These practices help improve transparency, accountability, and trustworthiness in AI systems.

Conclusion

A Deep Learning based Fake News Detection system was developed using an LSTM neural network.

The model achieved a test accuracy of 98.15%.

Responsible AI principles were considered through explainability, bias analysis, and mitigation recommendations.

This project demonstrates how AI systems can be made more transparent and trustworthy.

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